Task - 9: Network Vulnerability Scanning

Internship Task – Elevate Labs

This task focused on identifying network-level weaknesses using vulnerability scanning techniques. The main objective was to understand how open ports and exposed services increase the attack surface and how scanning tools help in early detection of security risks.

1. Introduction to Network Vulnerability Scanning

Network vulnerability scanning is the process of discovering open ports, running services, and potential weaknesses in networked systems. It is a key step in network security assessment.

2. Port Scanning

Port scanning was performed to identify open and closed ports on devices within the local network. Open ports indicate services that are accessible and may be targeted by attackers.

3. Using Nmap

Nmap was used as the primary scanning tool. It helps in discovering live hosts, identifying open ports, and detecting running services along with their versions.

4. Service Detection and Enumeration

Service detection was carried out to understand which applications are running on open ports. This information helps in identifying outdated or vulnerable services.

5. Operating System Identification

Nmap was also used to identify the operating system of target hosts. OS detection helps attackers and defenders understand system behavior and potential weaknesses.

6. Vulnerability Analysis

Based on open ports and detected services, potential vulnerabilities were analyzed. Unnecessary or misconfigured services increase security risks.

7. Interpreting Risks

Scan results were interpreted to assess the severity of identified risks. Services exposed to the network without protection can lead to unauthorized access.

8. Documentation of Findings

All scan results and observations were documented clearly. Proper documentation helps system administrators take corrective actions.

Learning Outcome

Through this task, I gained practical knowledge of network reconnaissance and vulnerability scanning. I learned how tools like Nmap help identify exposed services and why regular scanning is important for maintaining network security.